

Aim

To implement the Repository Pattern for data handling and use WorkManager to perform background tasks in an Android application.

Objectives

- Understand Repository Pattern in Android architecture
- Perform background tasks using WorkManager
- Separate UI logic from data handling logic
- Execute reliable background operations

Repository Pattern

- Acts as a **single source of data**
- Separates **UI layer** from **data operations**
- Improves code maintainability and testability

WorkManager

- Jetpack library for **background tasks**
- Works even if:
 - App is closed
 - Device restarts
- Used for:
 - Syncing data
 - Upload/download tasks
 - Periodic work

Procedure

1. Create a new Android project (Empty Views Activity, Kotlin).
2. Add WorkManager dependency in Gradle.
3. Create a Worker class for background task execution.
4. Create a Repository class to manage WorkManager.
5. Design UI with a Button.
6. Call Repository from MainActivity.
7. Run the app and check Logcat output.

Dependency (build.gradle)

```
implementation "androidx.work:work-runtime-ktx:2.9.0"
```

Worker Class

DataSyncWorker.kt

```
import android.content.Context
import androidx.work.Worker
import androidx.work.WorkerParameters

class DataSyncWorker(
    context: Context,
    params: WorkerParameters
) : Worker(context, params) {

    override fun doWork(): Result {

        // Background task simulation
        println("Data syncing in background using WorkManager")

        return Result.success()
    }
}
```

Repository Class

DataRepository.kt

```
import android.content.Context
import androidx.work.OneTimeWorkRequestBuilder
import androidx.work.WorkManager

class DataRepository(private val context: Context) {

    fun syncData() {
        val workRequest = OneTimeWorkRequestBuilder<DataSyncWorker>()
            .build()

        WorkManager.getInstance(context).enqueue(workRequest)
    }
}
```

UI Layout

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
```

```
<Button
    android:id="@+id/btnSync"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Sync Data"/>
</LinearLayout>
```

MainActivity.kt

```
import android.os.Bundle
import android.widget.Button
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val repository = DataRepository(this)
        val button = findViewById<Button>(R.id.btnSync)

        button.setOnClickListener {
            repository.syncData()
        }
    }
}
```

Output

- User clicks **Sync Data** button
- WorkManager triggers background task
- Message appears in **Logcat**:

Data syncing in background using WorkManager

